SCIENCE DEPARTMENT • SCIENCE LITERACY & LESSON INTERNALIZATION

PD AGENDA

Teaching Students to Analyze Science Texts With CUBS + CER

Science Literacy Learning Cycle — Learn and rehearse student-facing routines that help students analyze a science article, text, visual, data display, or model before constructing an evidence-based response.

WEDNESDAY, OCTOBER 28, 2026 • 2:30–3:30 P.M. **60 MINUTES** **GRADES 9–12 SCIENCE • BRING FORMS & EVIDENCE**

TODAY'S OUTCOME Leave with one ready-to-teach course-specific CUBS/CER routine, student directions, a model or anchor, a check for understanding, and a plan to collect completed writing for Friday's feedback and mastery work.

WHY THIS PLC MATTERS Students need explicit support to read, analyze, select evidence, and explain scientific thinking. CUBS and color-coded CER make these moves visible while keeping students responsible for the analysis and response.

PROTOCOL CONNECTION
Every Day Questions 1, 3, and 4; Team Deep Dive Steps 2–4: make the task and mastery bar visible, plan for anticipated responses and access, and design the student experience and facilitation moves.

WHAT TO BRING
One course-specific science article, text set, data display, model, investigation prompt, or constructed-response task; current standard(s); student-facing materials or a draft; highlighters in CER colors; Lesson Internalization Tool; and laptop/planning materials.

FOCUS TO PROTECT
The shared Biology benchmark is a model. Every science teacher adapts CUBS and color-coded CER to an authentic task in their course. The routine supports students in doing the task; it is not an additional grading system.

TODAY'S PLC PLAN

TIME (MIN)	FOCUS	WHAT TO DO	PRODUCT / CHECK
2:30–2:35	**Launch the strategy work**	Connect Monday's benchmark experience and Tuesday's planning to student-facing science literacy routines across all courses.	**Purpose**
2:35–2:41	**See the full instructional cycle**	Trace the path from experiencing the task to planning learning, teaching CUBS/CER, reading completed writing, providing feedback, and responding to evidence.	**Learning arc**
2:41–2:50	**Model CUBS**	Think aloud how students circle the main idea, underline supporting details, box high-leverage vocabulary, and summarize/record questions.	**Source analysis**
2:50–3:00	**Practice CUBS on a science source**	Use the Biology food web/Question 5 or another shared visual/text to identify what students need to notice before they write.	**Practice**
3:00–3:08	**Model color-coded CER**	Make claim, evidence, reasoning, and linking language visible in a science response; identify how students can check the structure of their own work.	**Response analysis**
3:08–3:15	**Teach the routine without giving answers**	Plan modeling, prompts, practice, feedback, and revision that keep students responsible for the scientific explanation.	**Facilitation**
3:15–3:28	**Protected course-alike application**	Adapt CUBS/CER to one authentic course task. Write student directions, a model/anchor, and a check for understanding or written-feedback opportunity.	**Ready-to-teach plan**
3:28–3:30	**Close + collect writing**	Teach the routine, collect a small de-identified set of completed writing, and bring it Friday for written-feedback and mastery analysis.	**Friday bridge**

PROTECTED INDIVIDUAL WORK WITH COLLEAGUE SUPPORT

WORK WITH COLLEAGUES

WHAT TO DO Draft the student-facing routine independently first. Then use colleagues to check that the source/task, directions, model, student action, and feedback/check-for-understanding preserve the intended science and literacy demand.

1. **Model CUBS:** Make source-analysis moves visible before students write.
2. **Practice on a source:** Identify what a reader needs to notice, question, and use as evidence.
3. **Model CER colors:** Help students see the claim, evidence, reasoning, and transitions in a response.
4. **Adapt to your course:** Write ready-to-use directions, an anchor, and a check for understanding.
5. **Collect completed writing:** Use evidence Friday to provide feedback and plan a mastery next step.

USE THIS REFERENCE DURING GROUP WORK

CUBS MAKES SOURCE ANALYSIS VISIBLE
Circle the main idea; underline supporting details; box unfamiliar or important words; and summarize the science idea or record questions before writing.

COLOR-CODED CER MAKES RESPONSE THINKING VISIBLE
Students label their claim, evidence, and reasoning so they can see what is present, partial, missing, or not yet connected to the science idea.

USE THE ROUTINE TO SUPPORT THE TASK
Model the routine, prompt student analysis, and use feedback that leads to a revision. Do not complete the science thinking or create an unrelated activity.

COLLABORATION IN PRACTICE Protect time for thoughtful individual work; use respectful, student-centered feedback to make the next decision clearer.
TEACHER RUBRIC CONNECTION **Standard I: I-A-4 Well-Structured Lessons • Standard II: II-A-2 Student Engagement • Standard II: II-D-2 High Expectations • Standard IV: IV-C-1 Professional Collaboration**

